

PelletPirate V2 Controller — User's Manual

Firmware: esp32-v2 branch, build 9be74ae or later **Applies to:** PelletPirate V2 control board (ESP32-DevKitC), 3.5" LCD + rotary encoder, web dashboard **Document revision:** 1.0 — July 2026

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1. Important Safety Information

Read this section before operating the grill.

- **This controller operates a live fire appliance.** Never leave a burning grill unattended for extended periods without phone alerts configured (Section 13) and never rely solely on any automatic system to prevent a fire. The built-in safety systems (Section 16) reduce risk; they do not eliminate it.
 - **Mains voltage is present** inside the controller enclosure (fan, auger, and igniter circuits). Never open the enclosure while it is plugged in.
 - **The igniter element gets extremely hot.** The controller automatically disables it above 115 °F grill temperature, but treat the firepot as hot at all times during and after a cook.
 - **Shutdown mode exists for a reason.** Always end a cook with SHUTDOWN rather than pulling power. Shutdown runs the fan for 15 minutes with no new fuel, burning off pellets remaining in the firepot. Cutting power with a live fire leaves smoldering pellets without airflow.
 - **Use PT100 probes only.** See Section 4 — a PT1000 probe physically cannot be read by this controller and will show as a fault.
 - **Empty hopper hazard:** running the hopper dry mid-cook causes a flame-out. The controller detects and safely shuts down (Section 16), but check pellet levels before overnight cooks.
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2. Meet Your Controller

The PelletPirate V2 is a pellet grill controller with two complete user interfaces:

- **The panel:** a 3.5" color LCD with a click-and-turn rotary encoder. Everything needed to run a cook is available here, with no phone required — designed for greasy-hands operation at the grill.
- **The web dashboard:** every phone, tablet, or computer on your Wi-Fi network can open the controller's web page. The dashboard mirrors the panel completely and adds conveniences that suit a screen: tap-to-type temperatures, live graphs, cook log downloads, program authoring, and firmware updates. Most owners find the web dashboard becomes their daily driver, with the panel as the at-the-grill fallback.

The two interfaces are always in sync: a target changed on the phone appears on the LCD within a second, and vice versa.

Finding the web dashboard: once Wi-Fi is configured (Section 14), browse to the controller's IP address, shown on the panel under **Settings** → **WI-FI** (e.g., <http://192.168.1.178>). On Apple devices, <http://pelletpirate.local> usually works too.

What's plugged into it

Connection	Purpose
Probe jacks P1–P5 (3.5 mm)	Five PT100 temperature probes: one for the pit (grill) temperature, four for meat
Fan	Combustion air blower
Auger	Pellet feed motor
Igniter	Hot-rod ignition element
Mains inlet	120 V AC power

One probe jack is designated the **grill probe** — it measures pit temperature and drives all temperature control and safety logic. By default this is jack **P1**, but it is assignable (Section 12.4). The remaining four jacks are **meat probes 1–4**.

3. Quick Start — Your First Cook

1. **Fill the hopper** with pellets. Check it again if the last cook was long.
2. **Plug in the grill probe** (jack P1 by default) and clip it to the grate. Plug meat probes into any of the other jacks and insert them into the meat.
3. **Set up your probes** (either UI): give each meat probe its meat type, a **goal** temperature (when it's done), and optionally an **alarm** temperature with an action (e.g., "Wrap at 165").
4. **Set the grill target** (e.g., 225 °F).
5. **Start:** on the panel, choose **Start Now – Ignite** from the main menu and press **Start Ignite**. On the web, tap **START · IGNITE**. The fan starts, then the auger, then the igniter (staggered start).
6. **Ignition happens automatically.** When the pit passes 115 °F the controller declares the fire lit and switches itself to **Cook** mode, holding your target with its PID control loop.
7. **Cook.** Watch from the couch. Change targets any time. Alarms and goals will tell you when to act (and buzz your phone if Section 13 is set up).
8. **Finish:** when the meat is done, select **SHUTDOWN**. The auger stops, the fan runs 15 minutes to burn off remaining pellets, and the grill turns itself Off.

Tip: For the classic low-and-slow flavor sequence, start in **Super Smoke** while the meat is cold (it absorbs the most smoke below ~140 °F internal), then switch to **Cook** for the remainder. Or let a **Cook Program** (Section 9) run the whole sequence for you.

4. Temperature Probes

4.1 Probe type — PT100 only

The controller is built for **3-wire PT100 RTD probes** (100 Ω at 32 °F). This is a hardware property of the measurement circuit and cannot be changed in software.

⚠ PT1000 probes do not work. They look identical and fit the jack, but read ~1,100 Ω at room temperature — beyond the circuit's measurement ceiling. A PT1000 shows as a fault (0 °F / "--") or, worse, occasionally produces a plausible-looking wrong temperature during partial contact. **Check the listing when buying probes: it must say PT100.** Verify with a multimeter: a PT100 reads **108–110 Ω** across tip-to-sleeve at room temperature; a PT1000 reads ~1,080–1,120 Ω .

4.2 Wiring a probe plug (3.5 mm TRS)

3-wire PT100 probes have two wires of one color (commonly red) and one of another (commonly white). The two same-colored wires connect to the same end of the sensing element.

Plug contact	Wire
Tip	Red #1 (either red)
Ring	Red #2 (the other red)
Sleeve (barrel/body lug)	White

The two reds are interchangeable. **Verify before soldering:** the two same-end wires read ~0–1 Ω between each other; either of them to the third wire reads ~108–110 Ω at room temperature. Whichever pair reads near zero goes to tip + ring.

4.3 Probe care

- **Moisture is the #1 probe killer.** After long cooks, grease and condensation can wick into the probe sheath or the jack, causing erratic readings or fault (0 °F). A wet probe can often be recovered by drying it in a kitchen oven at ~250 °F for 2 hours.
- Keep the jack area clean; a drop of isopropyl alcohol on a plug cleans oxidation and grease.
- Seat plugs fully — a half-inserted plug reads as a short.
- Avoid hot-plugging probes mid-cook when practical; if you must, a brief FAULT/OK blip in the log is normal.
- Don't let probe cables touch the firepot or slide under the grease tray.

4.4 Goals vs. alarms — two different things

Each meat probe has two independent temperature triggers:

- **Goal** — the done temperature (e.g., brisket 203 °F). Drives the time-remaining estimate, the progress bar, and the graphs' dashed goal lines. When crossed, a **GREEN banner** announces the meat is done (and pushes to your phone).
- **Alarm** — an *action* temperature with a label (e.g., **Wrap** at 165 °F, **Baste**, **Beer Me**). When crossed, a **RED banner** tells you to go do the thing.

A brisket typically has both: alarm "Wrap" at 165 on the way to a goal of 203. Both alerts are **one-shot**: once you acknowledge them, they stay quiet for the rest of the cook (pulling the probe out to wrap the meat will not re-ring an alarm you already handled). Every new cook re-arms them fresh.

5. The Controls

5.1 The rotary encoder (panel)

Gesture	Action
Rotate	Move focus between items / change a value being edited
Click (short press)	Select the focused item / confirm a value
Long press (~1 s)	Context back-out (e.g., dashboard → main menu)
Hold ~1.2 s while an alert banner is showing	Acknowledge (silence) the alert

Notes on the alert hold: while a banner is on screen and the button is held, the encoder belongs entirely to the acknowledge gesture — rotation is ignored and the button release does not "click through" into whatever was behind the banner. After the banner clears there is a ~1.5 second settle period before the encoder resumes normal control.

5.2 Screen dimming

After **5 minutes** without encoder input, the LCD dims to 25 % brightness. Any encoder touch restores full brightness — that first touch only wakes the screen and is not passed through as a command. The display never turns fully off; it stays glanceable from across the patio.

5.3 The web dashboard

- **Target temperature:** tap **+5/-5**, or tap the number itself to type a value (100–499 °F).
 - **Mode chips:** the row of mode buttons below the target. Chips are dimmed when not applicable (e.g., you can't jump straight from Off to Cook — ignite first).
 - **Probe cards:** tap any probe card to open its setup editor (goal, meat type, alarm, alarm action).
 - **Alert banner:** appears at the top — red or green. **Tap it to silence.**
 - Everything else on the page — graphs, logs, programs, profiles, phone alerts, Wi-Fi, firmware — is described in its own section below.
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6. Cook Modes Explained

Mode	What the controller does	When to use it
Off	All outputs off.	Between cooks.
Ignite (Start)	Fan on; auger feeds at a fixed starting cycle; igniter element on. Automatically promotes itself to Cook the moment the pit passes 115 °F. If the grill is <i>already</i> above 115 °F (e.g., restarting after a power blip), pressing Start skips ignition entirely and goes straight to Cook — the button reads Resume Cook .	Beginning a cook; restarting a warm grill.
Smoke	Feast/famine smoke cycle — see below.	Maximum smoke flavor phase.
Super Smoke	A deeper feast/famine cycle: longer pauses, more smolder, more smoke, lower and swingier temperature.	The heaviest smoke, typically the first hours of a big cut.
Cook	Full PID temperature control to your target. The auger duty adjusts every 20 s; the fan modulates in gentle 2-second bursts when the pit is within ± 10 °F of target, scaling air with fuel.	The workhorse mode — holding any temperature 100–499 °F.
Keep Warm	Same control loop as Cook; conventionally used at a low target (~165 °F) to hold finished food.	Resting/holding after the meat is done.
Shutdown	Auger off (no new fuel), fan on for a 15-minute pellet burn-off, then Off.	Always end cooks this way.
Re-Ignite	Manual re-ignition sequence (same behavior as Ignite).	Relighting after a flame-out.

About Smoke and Super Smoke — read this once

Smoke modes are **not temperature-controlled the way Cook is**, and that's deliberate. Smoke flavor is produced when fresh pellets land on dying embers and *smolder* before catching fire; a steadily-burning pellet fire is too efficient and burns the flavor compounds up. Smoke modes therefore cycle the fire on purpose: feed a burst of pellets, let the fire fall back to embers, feed again. The oscillating temperature *is* the smoke machine.

What the target means in smoke modes: **a band center, not a setpoint**. The controller nudges the pause length (one step every 90 seconds) to keep the pit inside roughly **target ± 15 °F** while preserving the smolder cycle. Practical guidance:

- For classic heavy-smoke temperatures (150–180 °F), **set the target low — around 170 °F** — before entering a smoke mode.
- Setting a higher target (e.g., 225) in Smoke mode gives you a warmer, lighter-smoke compromise.
- If you want a *held* temperature, use **Cook** — that's what the PID is for. Smoke modes will always wander within their band.

Meat takes on most of its smoke early, while the surface is cold and moist (smoke ring formation stops around 140 °F internal). Spending the first 2–4 hours in a smoke mode and the rest in Cook captures most of the available benefit.

7. The Dashboards

7.1 The LCD cook dashboard

Top to bottom:

- **Cook Mode** (top-left) — the current mode.
- **FAN AUG IGN** (top-right) — live actuator status. Green = running (igniter shows red when energized), gray = off. The auger indicator blinking on and off during Cook is normal — that's the duty cycle.
- **CURRENT** — pit temperature, large.
- **TARGET** — the setpoint. Click it (when focused) to adjust in $\pm 5^\circ$ steps; click again to confirm.
- **Probe rows** (up to 4): number + meat type, current temperature, the alarm (red) and goal (green) temperatures, and a progress bar showing elapsed time (ET) vs. estimated time remaining (EST). Click a probe row to open its setup.
- **Bottom-right:**
 - **Next: P2 0:45** — the *soonest upcoming goal* across all probes: which meat finishes next and roughly when.
 - **Start 7/17 11:19 AM** — when this cook began. This timestamp is fixed at ignition and never changes with mode switches — total cook duration is always Start → Shutdown. It even survives power outages.

About EST (estimated time remaining): it is a projection from the last hour's climb rate. During a **stall** (the hours-long plateau big cuts hit around 150–170 °F internal), a rate projection would produce absurd numbers, so the display honestly reads "**EST: stall**" instead. When the stall breaks, the estimate returns and tightens rapidly.

7.2 The web dashboard

The same information plus:

- **ET • Start** line under the mode, and the **Next goal** line.
 - **GRAPH** — a rolling one-hour chart of pit, target, and all probes, with dashed goal lines. Updates every 30 seconds.
 - **COOK PROGRAMS / COOK PROFILES / COOK LOGS / PHONE ALERTS / WI-FI** sections — Sections 9–14.
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8. Alarms and Alerts

All alerts appear as a **banner**: across the top of the LCD (blinking) and at the top of the web page. **Green banners are good news** (a goal was reached). **Red banners need you** (an action or a problem).

To silence: **hold the encoder button ~1.2 s**, or **tap the banner on the web page**. If phone alerts are configured, every one of these also buzzes your phone the moment it fires.

Alert	Color	Meaning	After you acknowledge
P2 Pork Butt 165°F – Wrap!	Red	A meat probe hit its action alarm.	One-shot: stays quiet for the rest of this cook.
Brisket DONE! P1 at 203°F	Green	A meat probe reached its goal.	One-shot: stays quiet for the rest of this cook.
GRILL TEMP DROP	Red	The pit fell 30 °F+ below target for over a minute <i>after</i> having reached temperature. Wind, an open lid, or a dying fire.	Re-arms automatically once the pit recovers into band. Clears itself if the pit recovers before you even get there, and also clears if it fired only because you raised the target (a raise is a climb, not a drop).
CHECK PELLETS – auger max, temp falling	Red	Fuel starvation signature: the auger is running flat-out while the pit dives. Empty hopper, bridged pellets, or a jammed auger. You have roughly 10 minutes before the flame-out shutdown takes over.	Re-arms if the condition returns.
GRILL SENSOR FAULT – SHUTTING DOWN	Red	The pit probe went dead mid-cook; the controller is blind and has forced a shutdown burn-off.	Investigate the grill probe/jack.
WRAP NOW (or other gate text)	Red	A running Cook Program has reached a human step and is holding until you act. Acknowledging it tells the program to continue.	Program advances to the next step.
BRISKET DONE (program completion)	Green	A Cook Program reached its final holding step.	Informational.
PROFILE HOLD – driver probe fault	Red	A running program lost its driver probe signal and is holding the current step.	Clears automatically when the probe returns.

Alarm hygiene built into the firmware, so you don't have to think about it: probe alarms and goals **stand down automatically outside of cooking modes** — probes dangling in hot pit air during Shutdown will not ring alarms for meat that's already on the cutting board.

9. Cook Programs (Automated Cooking)

A **Cook Program** runs an entire cook for you as a sequence of steps: *"Super Smoke until the brisket hits 140. Then Cook at 225 until 160. Then tell me to wrap and wait. Then Cook at 250 until 203. Then hold warm and tell me it's done."*

9.1 The model: one driver, many passengers

The grill has one fire, so a program follows exactly **one probe — the driver** (you pick which when you start it; usually the most important meat on the grill). Every temperature step in the program reads the driver.

All other probes are **passengers**: their own goals and alarms keep working normally (green DONE banners, wrap alarms, phone pushes), but they never influence the pit. This matches how pitmasters actually run mixed cooks — the brisket gets the pit it wants; the chickens come off when their alarm says so.

9.2 Running a program

On the web dashboard, **COOK PROGRAMS** section:

1. Tap a program (★ marks the built-in factory programs).
2. Choose the **driver probe** (1–4).
3. Confirm. If the grill is Off, the program ignites it first; if it's already running, the program takes over from the current state.

The status card shows the program name, driver, and current step (e.g., *"Step 2/5: cook 225 until P1 ≥ 160"*). Every step transition is recorded in the cook log and pushed to your phone.

Gates: when the program reaches a human step (wrapping, saucing), it raises a red banner with your instruction and **waits** — the pit holds its current behavior, nothing advances until you acknowledge (on either UI, or from the phone by opening the dashboard). Do the deed, silence the banner, and the program moves on. Probe readings dipping while you handle the meat cannot confuse it — a step once completed stays completed.

You always outrank the program. If you manually change the mode or target mid-program, the program notices and **pauses** itself, showing "PAUSED (manual control)". Tap **RESUME** to hand control back (it re-asserts the current step), or **STOP PROGRAM** to dismiss it — the pit simply stays in whatever mode it's in. Turning the grill Off or starting Shutdown always ends the program.

Power outages: a running program survives them. When power returns and the grill auto-resumes the cook (Section 17), the program picks up at the same step. Time-based steps restart their clock; temperature steps just keep watching the driver.

9.3 The factory programs

Four ★ factory programs ship with the firmware. They're read-only — open one with **VIEW**, then save it under a new name to customize.

- ★ **Brisket_Boss** — the canonical low-and-slow journey: 1. Super Smoke until driver ≥ 140 °F (maximum smoke while the meat can take it) 2. Cook @ 225 until driver ≥ 160 3. **GATE: WRAP NOW** (butcher paper — acknowledge to continue) 4. Cook @ 250 until driver ≥ 203 5. Keep Warm @ 165, green **BRISKET DONE** — holds until you're ready
- ★ **PorkButt_Solo** — same shape, more forgiving numbers: 1. Super Smoke until ≥ 150 · 2. Cook @ 250 until ≥ 165 · 3. GATE: WRAP · 4. Cook @ 275 until ≥ 203 · 5. Keep Warm + PORK DONE
- ★ **Ribs_Champ** — the 3-2-1 clock cook (ribs are too thin for probe temps, so the steps run on time): 1. Smoke for 3 hours · 2. GATE: WRAP + LIQUID · 3. Cook @ 225 for 2 hours · 4. GATE: UNWRAP + SAUCE · 5. Cook @ 250 for 1 hour · 6. Green **RIBS DONE – check bend** (no hold)
- ★ **Chicken_Crispy** — hot and fast, skin first: 1. Smoke for 45 minutes (all the smoke poultry needs) · 2. Cook @ 375 until driver ≥ 165 · 3. Green **PULL NOW** — deliberately no Keep Warm, because holding steams the skin soft.

9.4 Writing your own programs

Tap **NEW PROGRAM** (or VIEW a factory one and rename). A program is a few plain-text lines:

```
ver=1
step=supersmoke,0,temp,140,
step=cook,225,temp,160,
step=gate,0,ack,0,WRAP NOW
step=cook,250,temp,203,
step=keepwarm,165,end,0,DONE
```

Each `step=` line is `mode,target,exit,value,note` :

Field	Options	Meaning
mode	<code>supersmoke</code> <code>smoke</code> <code>cook</code> <code>keepwarm</code> <code>gate</code>	What the pit does. <code>gate</code> leaves the pit as-is and raises the note as a banner.
target	°F (0 for gates; band center for smoke modes)	Pit setpoint for this step.
exit	<code>temp</code> <code>time</code> <code>ack</code> <code>end</code>	What ends the step: driver reaching a temperature, elapsed minutes, your acknowledgment (gates only), or never (<code>end</code> = terminal hold).
value	°F for <code>temp</code> , minutes for <code>time</code> , 0 otherwise	The exit threshold.
note	free text	Banner text for gates and terminal steps; shows in logs.

Rules enforced by the controller: up to 8 steps; `gate` steps must use `ack` (and only they may); temperature exits 50–400 °F; time exits 1–1440 minutes. A program that parses to zero valid steps is rejected at save.

Program files live on the controller's storage and **survive firmware updates**. They are deleted by saving an empty body under the same name.

10. Cook Profiles (Saved Setups)

Separate from programs: a **profile** is a snapshot of your *setup* — grill target plus all four probes' meat types, goals, and alarms. Save the current setup with **SAVE CURRENT COOK**, and load it before a future cook to restore the whole configuration in one tap. Loading a profile does not start the grill or run any sequence — it just fills in the settings. (Think of a profile as *what* you're cooking; a program as *how the pit behaves over time*.)

Profiles can be saved/loaded/deleted from both UIs (LCD: **COOK PROFILES** in the main menu).

11. Cook Logs and Graphs

The controller keeps a flight recorder for every cook.

- **One CSV file per cook**, named by start time (`cook_20260717_1603.csv`), created when the grill leaves Off and closed when it returns.
- A **sample row** is written every 30 s (or 10 s — Section 12.2) with the full state: pit temp, target, control output, fan/auger/igniter state, all probe temps, Wi-Fi signal.
- **Event rows** are written immediately for everything notable, with the source attributed: mode changes (`lcd / web / auto`), target changes, probe configuration, alarms, goals, program steps and gates, probe faults and recoveries, safety actions.
- **Every row is committed to flash the moment it's written.** A power cut loses at most the last 30 seconds — and when the cook auto- resumes after an outage, logging **continues in the same file** with a `POWER LOSS - resumed` marker, so one cook stays one file.

Viewing: the web page's **COOK LOGS** section lists all logs with a **GRAPH** button (full-cook chart with goal lines) and a download link (the CSV opens directly in any spreadsheet). The **GRAPHS** item on the LCD main menu shows the live rolling hour.

Capacity: storage holds roughly 6–7 long cooks; when space runs low the oldest logs are deleted automatically. Download anything you want to keep forever.

12. Settings Reference

Panel: **Main Menu** → **SETTINGS**. Each row is focused by rotating and activated by clicking.

12.1 BRIGHTNESS

Display backlight level, **10–100 %** in 10 % steps. Click the row to enter adjustment (value turns green), rotate to change — the backlight updates live as you turn — and click again to save. The saved level persists across reboots and is the level the screen returns to when it wakes from idle dimming (Section 5.2).

12.2 COOK LOG

Sampling interval for the cook log. Clicking cycles:

- **30s** (default) — one sample every 30 seconds. A 14-hour cook is about 120 KB. Right for almost everyone.
- **10s** — three times the detail, three times the file size. Useful when studying PID behavior.
- **OFF** — no cook files are created at all (events included).

This setting only affects logging; it changes nothing about control.

12.3 WI-FI (status) and WI-FI SETUP

The status block shows the connected network, signal strength, and the dashboard address — this is where you look up the controller's IP. **WI-FI SETUP** scans for networks and lets you join one using the encoder (see Section 14 for the full Wi-Fi story, including phone-based setup).

12.4 GRILL PROBE JACK

Selects **which physical probe jack (P1–P5) carries the pit/grill probe**. All five jacks are electrically identical; this setting is purely an assignment. The remaining four jacks become meat probes 1–4, numbered in ascending jack order.

Clicking the row cycles P1 → P2 → ... → P5. Two things to know:

- **The change is only accepted while the grill is Off.** The grill probe drives the PID, the igniter lockout, and the sensor-fault shutdown; reassigning it mid-fire will be refused (the value shows "(run)" as a reminder).
 - The default is **P1**. Most owners never need to change this; it exists so a damaged jack never ends a cookout — move the pit probe to any healthy jack and reassign.
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13. Phone Alerts

Every alarm, goal, program gate, and safety event can buzz your phone — strongly recommended for overnight cooks. The controller uses **ntfy.sh**, a free push service that requires no account.

One-time setup (about two minutes):

1. Install the **ntfy** app on your phone (App Store / Play Store), or just keep `https://ntfy.sh/your-topic` open in a browser.
2. Invent a **topic name** — treat it like a password, since anyone who knows it can see your alerts. Something like `pirate-grill-x7k2`. Letters, digits, `-` and `_` only.
3. In the app, **subscribe** to that topic.
4. On the web dashboard, **PHONE ALERTS** section: enter the same topic and tap **SAVE & TEST**. Your phone should buzz within seconds ("Notifications enabled — you're connected!").

That's it. From then on: *"P2 Pork Butt 165F - Wrap!"*, *"Brisket DONE! P1 at 203F"*, *"CHECK PELLETS - auger max, temp falling"*, *"WRAP NOW"* — wherever you are, including 4 AM. Clear the topic field and save to disable. Multiple phones can subscribe to the same topic; the whole family can follow the brisket.

14. Wi-Fi Setup

The controller joins your home Wi-Fi as a normal device.

First-time setup (no network configured): about 25 seconds after power-up, the controller raises its own open hotspot named **PelletPirate-Setup**. Join it with your phone, browse to `http://192.168.4.1`, tap **CHANGE WI-FI**, pick your network, enter the password. The controller joins your network and drops the hotspot; its new address appears on the panel under Settings → WI-FI.

From the panel: Settings → **WI-FI SETUP** scans networks; pick one with the encoder and enter the password on the on-screen keyboard (rotate to a key, click to type).

Changing networks later: either method works any time; credentials are stored and survive reboots and firmware updates.

Notes: the grill never *depends* on Wi-Fi — every control function works from the panel with the network down, and the controller reconnects automatically when the network returns. `http://pelletpirate.local` works from Apple devices; Android and Windows are happier with the IP address.

15. Firmware Updates

Updates install **over Wi-Fi** — no cables, no disassembly.

1. Obtain the new firmware file (`pelletpirate.bin`).
2. Make sure the grill is **Off** (an update reboots the controller).
3. Browse to `http://<controller-ip>/update` .
4. Choose the file and tap **Upload**. Progress shows as a percentage; at 100 % the controller flashes the new firmware and reboots. It's back on the network in about 30 seconds.

What survives an update: everything — Wi-Fi credentials, all settings, probe setups, saved profiles, your custom programs, and cook logs. Factory programs update *with* the firmware.

If an update goes wrong: the controller keeps the previous firmware in a second slot. A new firmware must run healthily for 60 seconds to become permanent; if it crashes before that, the next boot **automatically rolls back** to the previous version.

Updating mid-cook is refused by default. The page offers a "force while cooking" override: the controller reboots for ~30 seconds and relies on the auto-resume system (Section 17) to pick the cook back up. Use it only when a fix genuinely can't wait.

16. Built-In Safety Systems

These run always, in every mode, regardless of what any program or person asks for:

System	Behavior
Boot-safe outputs	Fan, auger, and igniter lines are driven OFF in the first instants of every boot, before anything else initializes.
Igniter lockout	The igniter element is physically inhibited whenever the pit is above 115 °F, in every mode.
Igniter timeout	If the igniter runs 20 minutes without the pit reaching 115 °F, the controller gives up and forces Shutdown (no endless glowing element over a pile of pellets).
Combustion air interlock	The auger and igniter can never run without the fan.
Staggered start	Fan, then auger (+2 s), then igniter (+4 s) — motor inrush currents never stack.
Sensor-fault shutdown	If the pit probe reads as dead for 30 s while running, the controller stops feeding pellets and forces a Shutdown burn-off — it will not fly blind.
Flame-out shutdown	After the pit has reached temperature: if it falls 60 °F+ below target and stays there 10 minutes (igniter assist having had its chance), the fire is presumed dead — forced Shutdown rather than piling fuel into a dead pot.
Pellet starvation warning	~10 minutes <i>before</i> flame-out would trigger: auger pegged + temperature diving raises CHECK PELLETS (and a phone push). A fast hopper refill can often save the fire.
Watchdog	If the control software ever hangs, the hardware watchdog reboots the controller within 10 s — and a reboot lands in boot-safe state.
Shutdown burn-off	15 minutes of fan-only operation empties the firepot of burning pellets before power-down.

17. Power Outages

The controller is designed so a power blip is a non-event:

- **While cooking, the running mode and target are saved continuously.** When power returns, the controller waits for a valid pit reading and then decides:
 - Pit still at/above 115 °F → **the cook resumes automatically** in the same mode at the same target. No buttons.
 - Power was lost during **Shutdown** → the burn-off **always resumes** (fan on, no fuel) regardless of temperature — safety first.
 - Pit has gone cold → the grill stays **Off**. A cold grill never relights itself unattended; that decision needs a human. Walk out and press **Resume Cook** (the ignite screen's hot-restart is not needed when cold — a normal Start Ignite relights it).
 - **The cook log continues in the same file** with a `POWER LOSS - resumed` marker; elapsed time and the Start timestamp carry through as if nothing happened.
 - **A running Cook Program resumes at its current step.**
 - If the outage killed the fire but left the pit warm (a long outage), the resumed cook may flame-out-shutdown shortly after — that's the safety net working. Relight manually after checking the firepot.
-

18. Troubleshooting

A probe reads 0 °F / "--" (fault): - Nothing plugged into that jack → normal, that's how empty reads. - Plug not fully seated → push it home. - New probe → is it actually a **PT100**? (Section 4.1 — a PT1000 reads as fault/garbage.) Check with a multimeter: ~109 Ω tip-to- sleeve at room temperature. - After a long/greasy cook → moisture in probe or jack. Clean the jack (isopropyl), dry the probe (oven, 250 °F, 2 h). - Reads fine in a different jack → the original jack needs cleaning; or reassign jacks (Section 12.4) and cook on.

Probe reads a plausible but wrong temperature: partially seated plug, moisture leakage (reads *low*), or wrong probe type making intermittent contact. Reseat, dry, verify type.

"GRILL TEMP DROP" nags during normal operation: it now requires a sustained (60 s+) drop and clears itself on recovery or on target raises. If it still fires often, your pit may genuinely be under-recovering — check pellet quality (moist pellets burn weak) and wind exposure.

Smoke mode won't hold the temperature I set: by design — see Section 6. Use Cook for held temperatures; set a low target (~170) in smoke modes for classic smoke-range temps.

EST says "stall": normal for big cuts between ~150–170 °F internal. Not a malfunction — the estimate resumes when the meat starts climbing again. (Wrapping shortens the stall.)

Web page unreachable: check Settings → WI-FI on the panel for the IP and connection status. Use the IP rather than `pelletpirate.local` on Android/Windows. If the controller lost your network it will retry forever on its own; if it was never configured, look for the **PelletPirate-Setup** hotspot (Section 14).

Phone alerts not arriving: confirm the app is subscribed to *exactly* the topic saved on the dashboard (SAVE & TEST should buzz); confirm the controller's network has internet access, not just LAN.

The fan sounds different / pulses: in-band fan modulation runs 2-second bursts by design. If your particular fan hums or surges objectionably, report it — a one-line firmware change restores continuous fan.

Alarm keeps re-ringing after I handled it: it shouldn't — probe action alarms and goals are one-shot per cook on this firmware. A *grill temp drop* or *check pellets* alarm re-arms by design because those are ongoing conditions, not actions.

The encoder did something I didn't intend right after silencing an alarm: it shouldn't on this firmware (the ack gesture is isolated and followed by a 1.5 s settle). If you can reproduce it, note what screen you were on.

19. Specifications

Item	Value
Grill target range	100–499 °F
Probe goal/alarm range	100–499 °F (0 = off)
Temperature probes	5 × 3-wire PT100 RTD, 3.5 mm TRS jacks
Pit control	PID, 20 s cycle, auger duty 15–100 %
Smoke cycle	15 s feed / 55–105 s pause, self-adjusting (target ± 15 °F band)
Fan modulation	2 s burst cycle within ± 10 °F of target; always-on otherwise
Ignite → Cook threshold	115 °F
Igniter safety timeout	20 minutes
Shutdown burn-off	15 minutes
Sensor-fault shutdown	after 30 s of dead pit probe
Flame-out shutdown	target -60 °F sustained 10 min (after reaching band)
Temp-drop alarm	target -30 °F sustained 60 s (after reaching band)
Starvation warning	auger ≥ 95 % + pit -20 °F in 4 min
Cook log	CSV, 30 s or 10 s samples, ~6–7 long cooks retained
Cook programs	up to 8 steps; temp/time/gate/terminal exits
Display idle dim	25 % after 5 min, wake on encoder
Alert acknowledge	hold encoder ~ 1.2 s / tap web banner
Connectivity	Wi-Fi 2.4 GHz; web dashboard + WebSocket live updates; ntfy.sh push
Firmware update	over-the-air via <code>/update</code> , automatic rollback on failed boot
Power-loss resume	automatic when pit ≥ 115 °F; Shutdown always resumes

PelletPirate V2 — built by hand, hardened by brisket.